Market Power, Mergers, Digital Area, and Competition

Focus on EU Policies and the Digital Markets Act (DMA)

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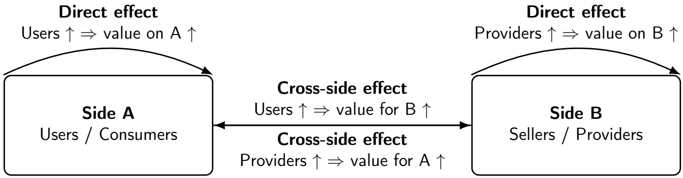
Introduction to the Digital Economy

- Profound transformations in the competitive landscape.
- Rise of tech giants and platform-based businesses.
- Reshaping of traditional industries and creation of new market dynamics.

Market Power in the Digital Era

- New forms in the digital context:
  - Network effects
  - Data-driven strategies
  - Digital monopolies

Direct and Cross-Side Network Effects (Platform View)



Positive feedback loop: growth on one side attracts the other side and can create tipping.

Mergers and Acquisitions in the Digital Context

- Potential concerns:
  - Entrenching dominance
  - Limiting market entry for smaller players
- Impact on competition and innovation.

Regulatory Challenges

- Balancing innovation and competition.
- Ensuring dynamic and fair markets.
- Adapting to rapid changes in the digital economy.

Course Objectives

- Understand key concepts of market power, mergers (traditional and digital markets), and competition.
- Analyze their implications in the digital economy.
- Explore regulatory approaches and challenges.

Course Overview

1. Fundamentals of Market Power
   - Definition
   - Sources of Market Power
   - Measurement Tools
   - Limitations
2. Merger Control and Policies
   - Different Forms of Mergers
   - Historical Movements
3. Key Industries and Examples
   - Recent Typologies of MA
   - Mergers That Enhance Economic Efficiency
   - Mergers Without Efficiency Gains
   - Motives for Mergers
   - Section I: First Economic Analyses of Horizontal Mergers
4. Amazon as a Digital Ecosystem
5. Core Pillars of the Amazon Ecosystem
6. Amazon Ecosystem: Visual Overview

Chapter I: Fundamentals of Market Power

Fundamentals of Market Power

- **What is Market Power?**
  - Market power refers to a firm's ability to influence price, output, or product quality.
  - Firms with market power act as *price makers*, unlike firms in perfectly competitive markets.
- **Economic intuition:**
  - Market power allows firms to set prices above marginal cost.
  - This leads to higher prices, reduced quantities, or lower quality for consumers.
  - However, market power can also finance innovation (e.g., R&D-intensive industries).
- **Welfare implications:**
  - Reduced consumer surplus.
  - Potential allocative and productive inefficiencies.

Structural Sources of Market Power

- **Economies of Scale**
  - Large firms spread fixed costs over higher output, lowering average costs.
  - Smaller competitors face higher unit costs and struggle to enter.
  - **Example:** Walmart benefits from large-scale purchasing and logistics optimization.
- **Network Effects**
  - The value of a product increases as more users adopt it.
  - Strong network effects create lock-in and high switching costs.
  - **Example:** Facebook becomes more valuable as its user base grows.
- **Product Differentiation**
  - Firms differentiate products through branding, design, or ecosystems.
  - Differentiation reduces price elasticity of demand.
  - **Example:** Apple justifies premium pricing through ecosystem integration.
- **Barriers to Entry**
  - Legal, technological, or financial obstacles prevent new competitors.
  - **Example:** Patents in the pharmaceutical industry.

Strategic and Regulatory Sources

- **Pricing Strategies (Predatory Pricing)**
  - Firms temporarily set prices below cost to eliminate rivals.
  - Difficult to prove empirically and legally.
  - **Example:** Amazon has faced accusations of predatory pricing.
- **Mergers and Acquisitions**
  - Reduce competition by consolidating market power.
  - **Example:** Facebook's acquisition of Instagram.
- **Government-Granted Monopolies**
  - Monopolies justified by efficiency or infrastructure needs.
  - **Example:** Electricity and water utilities.
- **Trade Protections and Patents**
  - Shield domestic firms from foreign competition.
  - **Example:** Boeing in the aerospace industry.

Concentration Ratios (CR)

- **CR4 / CR8:** Sum of market shares of the largest 4 or 8 firms.
- High ratios suggest potential market power.
- Used as a preliminary screening tool by competition authorities.
- **Example:** U.S. wireless telecom market CR4 exceeds 90%.

Herfindahl-Hirschman Index (HHI)

- **Definition:**

$$HHI = \sum_{i=1}^{n} s_i^2$$

- **Economic intuition:**
  - Squaring market shares gives more weight to dominant firms.
  - Market dominance matters more than fragmentation.
- **Interpretation:**
  - HHI $<$ 1,500: Competitive market.
  - 1,500 $\leq$ HHI $\leq$ 2,500: Moderately concentrated.
  - HHI $>$ 2,500: Highly concentrated.
- **Example:**
  - U.S. airline industry HHI $\approx 2,500$.

Calculating the HHI: Example

Market shares:

$$s_1 = 50\%, \quad s_2 = 30\%, \quad s_3 = 20\%$$

Calculation:

$$HHI = 50^2 + 30^2 + 20^2 = 3800$$

Interpretation:

- Highly concentrated market.
- Strong indication of market power.

Lerner Index and Price-Cost Margin

- **Lerner Index:**

$$L = \frac{P - MC}{P}$$

- **Numerical illustration:**
  - If $P = 100$ and $MC = 60$, then $L = 0.4$.
  - The firm applies a 40% markup over marginal cost.
- **Price-Cost Margin (PCM)**
  - Used when marginal cost is difficult to observe.
  - **Example:** Luxury brands such as Louis Vuitton.

HHI and the Lerner Index

- In a Cournot oligopoly:

$$HHI \approx \frac{L}{\epsilon}$$

- ϵ: Price elasticity of demand.
- **Insight:**
  - HHI is market-wide.
  - Lerner Index is firm-specific.

Limitations of the HHI

1. **Structural Limits**
   - Highly sensitive to market definition (product and geographic).
   - Ignores product differentiation and quality differences.
   - May overstate concentration in markets with differentiated products (e.g., automobiles).
2. **Dynamic Limits**
   - Static snapshot: does not capture innovation or future competition.
   - Ignores potential entry, technological disruption, and creative destruction.
   - High HHI may coexist with intense innovation (e.g., digital platforms).
3. **Data Limits**
   - Requires precise and comprehensive market share data.
   - Revenue-based shares may be misleading in zero-price markets.
   - User-based and data-based power are not captured.

Limitations of Market Power Measurement

- Traditional indicators focus on prices and quantities.
- Digital and multi-sided markets require complementary tools.
- Interpretation varies across regulatory frameworks.

Discussion: Is HHI Relevant in Digital Markets?

- Useful as a first diagnostic of concentration.
- High HHI often reflects strong network effects.
- Example: Search engines with dominant players.

Discussion: Limits in Digital Ecosystems

- Platforms operate across interconnected markets.
- Free services complicate revenue-based metrics.
- Market power may arise from data control, not market share alone.

Open Questions

- Should competition policy focus on prices, data, or innovation?
- Are traditional indicators sufficient for digital platforms?
- Should regulation be ex ante or ex post?

Chapter II: Merger Analysis and Competition Policy

Different Types of Mergers

- Mergers can be classified into several categories:
  - **Horizontal Concentration**
    - Combines activities at the same stage of production.
    - Most common form of merger.
    - **Example (Homogeneous)**: Disney acquiring Pixar.
    - **Example (Heterogeneous)**: Coca-Cola acquiring Costa Coffee.
  - **Vertical Acquisition**
    - Integration along the production chain (vertical integration).
    - Example: Tesla acquiring SolarCity to integrate energy generation and storage.
  - **Concentric Acquisition**
    - Often part of a diversification strategy.
    - Objective: Develop specific technology in a related market.
    - Example: Apple acquiring Beats Electronics to expand its music ecosystem.
  - **Conglomerate Acquisition**
    - Purely financial objective.
    - Aims to develop a diversified portfolio of businesses.
    - Example: Alphabet acquiring Fitbit for health data insights.

Historical Waves of Mergers in the U.S.

- **First Wave (1883-1904)**
  - Characterized by horizontal mergers involving numerous firms.
  - Objective: Establish monopolistic dominance.

Oil Industry

- The Standard Oil Trust (1882, later dissolved in 1911) was formed by John D. Rockefeller.
- Consolidated over 40 oil companies.
- Created a near-monopoly in the oil industry by controlling production, refining, and distribution.

Steel Industry

- In 1901, J.P. Morgan orchestrated the merger of Carnegie Steel, Federal Steel, and several other companies.
- Formed U.S. Steel Corporation, the first billion-dollar corporation.
- Dominated the steel market.

Railroads

- Consolidation of numerous small railroads into larger networks.
- Dominance of companies like the Pennsylvania Railroad and Union Pacific Railroad.
- Reduced competition and improved coordination of routes.

Tobacco Industry

- The American Tobacco Company was formed in 1890.
- Merged several smaller tobacco firms.
- Controlled over 90% of the U.S. tobacco market.

Sugar Industry

- The American Sugar Refining Company (Sugar Trust) was formed in 1887.
- Controlled most of the sugar refining capacity in the United States.

Utilities

- Consolidation in the electric power and gas sectors.
- Companies like General Electric (formed in 1892 through the merger of Edison General Electric and Thomson-Houston Electric Company) became dominant.

Driving Factors Behind Mergers

Improvements in production and transportation made economies of scale feasible. When marginal costs fall and fixed costs rise, firms seek scale. In IO terms: technology can change the cost structure and shift markets toward concentration.

Technological Advancements

- Improvements in production and transportation (e.g., railroads, steamships).
- Made economies of scale more feasible.

Legal and Policy Changes

- Weak enforcement of antitrust laws before the Sherman Antitrust Act of 1890.
- Allowed large-scale consolidation.

Economic Pressures

- Intense competition, falling prices, and overproduction.
- Forced many firms to merge to survive.

Financial Innovation

- Investment banks like J.P. Morgan & Co. facilitated and financed mergers.
- Enabled the creation of large corporations.

Antitrust Response

- Wave of mergers led to public outcry over monopolistic practices.
- Sherman Antitrust Act of 1890 was passed, but its enforcement was initially weak.
- Large trusts like Standard Oil and American Tobacco were only broken up later (1911).

Historical Waves of Mergers in the U.S. (Continued)

- **Second Wave (1916-1929)**
  - Strengthened the oligopolistic structure of U.S. markets.
  - Industries involved: Food, chemicals, and mining.
  - Goals:
    - Broaden product ranges.
    - Expand market shares.

Key Characteristics of the 1930s–1950s Merger Wave

- **Vertical Mergers:**
  - Dominated merger activity.
  - Companies integrated supply chains to control production and distribution.
- **Conglomerate Mergers:**
  - Diversification into unrelated industries to spread financial risks.
- **Limited Horizontal Mergers:**
  - Antitrust laws like the Sherman and Clayton Acts discouraged mergers between direct competitors.
- **Economic Context:**
  - Great Depression and WWII reduced merger activity.
  - Post-war period saw industrial growth and renewed merger trends.

Impact of the Celler-Kefauver Act (1950)

- **Closed the Asset Acquisition Loophole:**
  - Extended antitrust enforcement to asset purchases, preventing firms from bypassing regulation.
- **Regulation of Vertical and Conglomerate Mergers:**
  - Broadened antitrust scrutiny beyond horizontal mergers.
- **Strengthened Antitrust Policy:**
  - Enhanced the power of the FTC and DOJ to block anti-competitive mergers.
- **Market Effects:**
  - Reduced market concentration.
  - Helped protect small businesses and foster competition.

Summary of the Period (1930s–1950s)

- Merger activity focused on vertical integration and diversification.
- The Celler-Kefauver Act significantly shaped antitrust enforcement, reducing anti-competitive practices.
- Prepared the ground for a more balanced and competitive market environment.

Key Features of the Fourth Merger Wave in the U.S. (1980s–1990s)

- **Deregulation and Hostile Takeovers:**
  - Financial deregulation fueled merger activity.
  - Rise of leveraged buyouts and hostile takeovers.
- **Focus on Core Activities:**
  - Companies restructured to concentrate on core businesses.
  - Divestitures of non-core assets.
- **Internationalization:**
  - Growth of cross-border mergers.
  - U.S. firms sought global markets and synergies.
- **Technological Advancements:**
  - Innovations in telecommunications and computing facilitated mergers.

This wave also corresponds to a shift in antitrust philosophy in the US: more emphasis on price effects and efficiency claims.

Comparison with Europe

- **Market Structures:**
  - U.S.: More developed financial markets, facilitating mergers.
  - Europe: Fragmented markets, with regulatory barriers slowing cross-border mergers.
- **Regulatory Environment:**
  - U.S.: Lax antitrust enforcement during the 1980s encouraged consolidation.
  - Europe: Stricter competition policies under the European Commission.
- **Sectoral Focus:**
  - U.S.: Dominance in technology, finance, and media mergers.
  - Europe: Industrial and energy sectors saw significant consolidation.
- **Cultural and Strategic Differences:**
  - U.S.: Emphasis on shareholder value and short-term gains.
  - Europe: Focus on long-term industrial policy and national interests.

This comparison helps you understand why EU merger control is designed as a centralized system: to prevent a "race to the bottom" between Member States and to protect the single market.

Current Trends in Mergers and Acquisitions

- **Tech-Driven Mergers**
  - Aim: Acquire innovative technologies or enhance digital capabilities.
  - **Example**: Microsoft acquiring LinkedIn to expand professional networking and data integration.
- **Cross-Border Mergers**
  - Aim: Access new markets and diversify geographically.
  - **Example**: Tata Motors acquiring Jaguar Land Rover to expand into luxury vehicle markets.
- **Sustainability-Focused Mergers**
  - Aim: Align with environmental goals and reduce carbon footprint.
  - **Example**: Nestlé acquiring Sweet Earth Foods to enhance its plant-based product offerings.
- **Platform-Based Mergers**
  - Aim: Consolidate user bases and enhance network effects.
  - **Example**: Uber acquiring Postmates to strengthen its food delivery services.

In modern markets, mergers often target intangibles: data, algorithms, ecosystems, talent.

Motives for Mergers: Mergers That Enhance Economic Efficiency

- **Economies of Scale**
  - Objective: Achieve optimal size to minimize production costs.
  - Linked to fixed management costs and the indivisibility of fixed factors.
  - **Example**: Renault's stake in Nissan allowed them to combine their global distribution networks.

Motives for Mergers: Mergers That Enhance Economic Efficiency (pro-competitive)

- **Synergy Effects**
  - Economies of scope: $c_1(a) + c_2(b) > c(a + b)$.
  - Pursuit of industrial and financial synergies.
- **Acquisition of New Technologies**
- **Improvement in Company Management**
- **Overcoming Trade Barriers**
  - **Cross-border acquisitions and foreign direct investment (FDI)**.
- **Reduction in transaction costs (Coase, Williamson) and Production Costs**
  - Example: Vertical mergers.

Motives for Mergers: Mergers Without Efficiency Gains

- **Seeking Greater Market Power**
  - At the expense of competition and consumers.
- **Creating Barriers to Entry, lock in, foreclosure**
  - May lead to a dominant market position, risking abuse of dominance.
- **Increasing CEO Power**

Recent Examples and Trends in Mergers

- **Economies of Scale**: Amazon acquiring Whole Foods to optimize its supply chain and logistics.
- **Synergy Effects**: Disney acquiring 21st Century Fox to enhance its streaming content portfolio.
- **Technology Acquisition**: Google acquiring Fitbit to strengthen its position in wearable technology and health data.
- **Cross-Border Expansion**: Tata Motors acquiring Jaguar Land Rover to access luxury markets globally.
- **Vertical Integration**: Apple acquiring Intel's modem business to reduce dependency on external suppliers.

Vertical Mergers: Pro-competitive Effects

Efficiency-enhancing motivations

- **Reduction of transaction costs**
  - Avoid bargaining, monitoring, and renegotiation costs
  - Mitigate hold-up problems and opportunistic behavior
- **Elimination of double marginalization**
  - Coordination of upstream and downstream pricing decisions
  - Lower final prices and higher total surplus
- **Improved coordination and productive efficiency**
  - Better information flows along the value chain
  - Enhanced quality control and innovation incentives
- **Supply and demand security**
  - Securing access to critical inputs or distribution channels

Vertical integration can improve welfare by reducing inefficiencies arising from market transactions.

The Hold-Up Problem (Transaction Cost Economics)

Transaction Cost Economics highlights that market transactions are costly when contracts are incomplete and investments are relationship-specific.

Specific Investments and Contractual Incompleteness

- Hold-up arises when one party makes **specific investments** (physical, human, or site-specific assets).
- These investments are **sunk** and have low value outside the relationship.
- Contracts cannot specify all future contingencies.

Ex-post Renegotiation

- After the investment is made, the trading partner may attempt to:
  - Renegotiate contract terms,
  - Extract quasi-rents from the investor.
- This behavior weakens ex-ante investment incentives.

The Hold-Up Problem (Transaction Cost Economics)

Economic Consequences

- Underinvestment in relationship-specific assets,
- Inefficient contractual arrangements,
- Welfare losses.

Opportunism and Vertical Integration (Transaction Cost Economics)

Opportunistic Behavior

- Defined as *self-interest seeking with guile*.
- Takes various forms:
    - Strategic renegotiation,
    - Information withholding,
    - Threats of termination or delay.

When Opportunism is Most Severe

- High asset specificity,
- Uncertainty about future states of the world,
- Incomplete contracts.

Vertical Integration as a Governance Solution

- Replaces market transactions with hierarchical control.
- Limits ex-post bargaining and opportunistic threats.
- Secures investment incentives and reduces transaction costs.

Vertical Mergers: Anti-competitive Effects

Strategic and exclusionary motivations

- **Input or customer foreclosure**
  - Restrict rivals' access to essential inputs or consumers
  - Increase rivals' costs or limit their scale
- **Raising barriers to entry**
  - Vertical control may deter potential entrants
  - Reinforcement of incumbent advantages
- **Leveraging market power**
  - Extension of dominance from one stage of the value chain to another
- **Facilitation of collusion**
  - Increased information sharing may ease tacit coordination

Anti-competitive concerns arise when efficiency gains are outweighed by exclusionary incentives.

Section I. First Economic Analyses of Horizontal Mergers

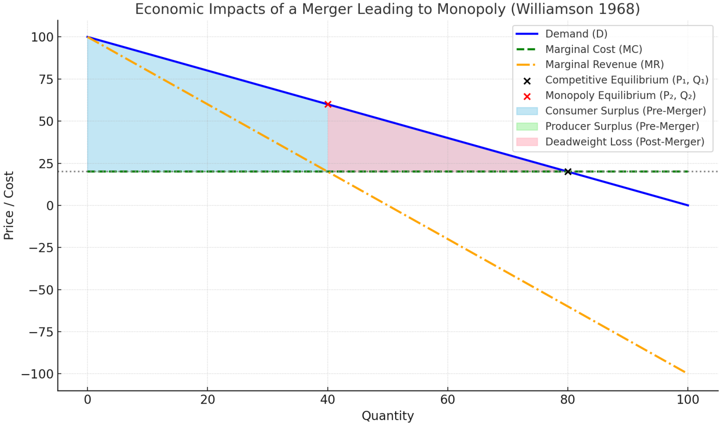
- Mergers can lead to the creation of monopolies.
- O. Williamson (1968) analyzed the economic trade-offs of such mergers.
- Focus: Balance between efficiency gains and welfare losses.

Key Concepts

- **Consumer Surplus**: Benefit consumers receive from paying less than they are willing to pay.
- **Producer Surplus**: Profit producers earn above their costs of production.
- **Deadweight Loss (DWL)**: Welfare lost due to reduced market efficiency.
- **Efficiency Gains**: Cost savings due to a reduction of the marginal cost or improved operations.

Pre-Merger vs. Post-Merger Equilibrium

- Pre-merger: Competitive market, lower prices, higher consumer surplus.
- Post-merger: Monopoly, higher prices, reduced consumer surplus, potential DWL.

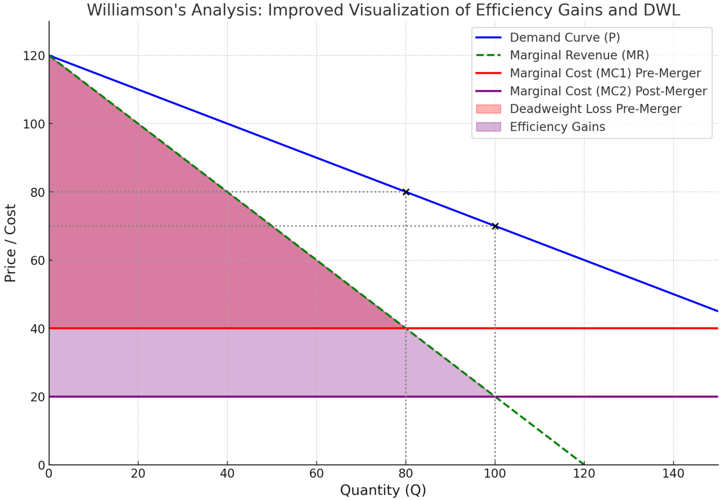


Williamson's Trade-Off

- A merger can result in both welfare losses and efficiency gains.
- Key question: Do the cost savings outweigh the welfare losses?

Efficiency Gains After the Merger

- Post-merger, marginal cost may decrease due to improved efficiency.
- Result: Lower prices, increased producer surplus, reduced DWL.



Conclusion

- Williamson's framework highlights the trade-off between market power and efficiency gains.
- Policy implications: Regulators must carefully evaluate mergers.
- Efficiency gains must significantly outweigh welfare losses to justify monopolistic mergers.
- Limitation: Don't take into account outsiders behavior!

Salant, Switzer et Reynolds (1983): Cournot Competition - Overview

Cournot Model Basics:

- Firms compete by choosing quantities.
- Market price is determined by the total quantity produced.
- Profit depends on the chosen quantity and the market price.

Key Assumptions:

- Homogeneous product.
- Linear inverse demand function: $P(Q) = a - bQ$, where $Q = q_1 + q_2 + q_3$.
- Cost function: $C_i(q_i) = cq_i$.

Objective: Each firm maximizes its profit:

$$\Pi_i = (P(Q) - c)q_i = (a - bQ - c)q_i$$

Pre-Merger Profits

Key Setup:

- Linear inverse demand: $P(Q) = a - bQ$, where $Q = q_1 + q_2 + q_3$.
- Marginal cost: c.
- Profit for Firm i:

$$\Pi_i = (P(Q) - c)q_i$$

Equilibrium Quantities:

$$q^* = \frac{a-c}{4b}, \quad Q^* = 3q^* = \frac{3(a-c)}{4b}$$

Pre-Merger Profits (Per Firm):

$$\Pi^* = \frac{(a-c)^2}{16b}$$

Total Profit Before Merger:

$$\Pi_{total}^{pre} = 3 \cdot \frac{(a-c)^2}{16b}$$

Post-Merger Profits (No Cost Reduction)

After Merger:

- Merged firm (A): Maximizes joint profit for two former firms 1 and 2.
- Independent firm (B): Maximizes its own profit.

Quantities:

- Merged firm produces: $q_A = \frac{a-c}{3b}$.
- Independent firm produces: $q_B = \frac{a-c}{3b}$.
- Total market quantity:

$$Q_m = q_A + q_B = 2\frac{a-c}{3b}.$$

Market Price:

$$P_m = a - bQ_m = a - \frac{b.2(a-c)}{3b} = \frac{a+2c}{3}.$$

Post-Merger Profits (No Cost Reduction)

Profits:

- Profit for merged firm:

$$\Pi_A = (P_m - c)q_A = \frac{(a-c)^2}{9b}.$$

- Profit for independent firm:

$$\Pi_B = (P_m - c)q_B = \frac{(a-c)^2}{9b}.$$

- Total profit after merger:

$$\Pi_{total}^{post} = \Pi_A + \Pi_B = \frac{2(a-c)^2}{9b}.$$

Comparison:

$$\Pi_A^{post} < \Pi_{1+2}^{pre}$$

Why Does Total Profit Decrease?

Key Factors:

- Merged firm produces less than the two firms did individually.
- Reduction in market output results in lower total revenue.
- Increased market power does not fully compensate for reduced output.

Consumer Welfare Impact:

- Higher prices harm consumers.
- Deadweight loss increases due to reduced total output.

Conclusion

Key Takeaways:

- Mergers in Cournot competition reduce total output and increase prices.
- Total profit may decrease compared to pre-merger levels.
- Consumer welfare declines, and regulatory oversight is necessary.
- Results could be different taking into account efficiency gains after the merger.

Key Insights from Perry and Porter (1985)

- They analyzed the competitive effects of horizontal mergers.
- Focus: Impact of mergers on market structure, pricing, and firm behavior, particularly in the presence of **cost asymmetries**.

Market Concentration and Pricing

- Increased market concentration typically leads to higher prices.
- Mergers reduce the number of competitors, softening price competition.
- Price effects depend on pre-merger market conditions and the market shares of the merging firms.

Capacity and Strategic Behavior

- Mergers may lead to reduced production capacity, further increasing prices.
- Capacity adjustments post-merger can be a strategic tool to enhance market power.

Impact of Cost Asymmetries

- They examined scenarios where merging firms have different cost structures.
- A merger between a high-cost and a low-cost firm may lead to:
  - Increased market power due to reduced competition.
  - Potential efficiency gains if the high-cost firm adopts the lower-cost firm's technologies or practices.
- However, these efficiency gains must be significant to offset potential price increases.

Profitability of Mergers with Asymmetric Costs

- Mergers between firms with asymmetric costs can be profitable if:
  - Cost efficiencies are realized post-merger.
  - The merged firm can leverage its lower combined cost structure to dominate the market.
- However, profitability may be limited if:
  - The cost disparity between the firms is too large to generate meaningful synergies.
  - Non-merging firms exploit the increased prices to gain a competitive edge (consistent with the "merger paradox").

Key Insights from Davidson and Deneckere (1985) on Mergers in a price Competition

- Davidson and Deneckere (1985) analyzed the effects of horizontal mergers in oligopolistic markets.
- Their work has been foundational in industrial organization and antitrust economics.

Impact on Market Prices

- Mergers reduce competition, leading to higher market prices.
- Even with cost efficiencies, the incentive for aggressive price competition decreases.

Merger Paradox

- Non-merging firms often benefit more than the merging firms.
- These firms capitalize on higher market prices without incurring merger costs.
- This phenomenon is referred to as the "merger paradox."

Coordination and Collusion

- Mergers can facilitate tacit collusion among remaining firms.
- Firms find it easier to coordinate pricing strategies post-merger.

Efficiency Gains vs. Market Power

- Mergers create a trade-off between efficiency gains and increased market power.
- Consumer benefits depend on whether cost reductions offset price increases.

Sequential and endogeneous Mergers

- Sequential merger models analyze how mergers unfold in multiple stages.
- Each merger influences market structure, profitability, and the likelihood of future mergers.

Stigler (1950s)

- Explored the role of mergers in stabilizing oligopolistic markets.
- Suggested that sequential mergers could mimic cartel behavior by reducing the number of competitors.
- Highlighted the risk of collusion and price increases due to reduced competition.

Kamien and Zang (1990)

- Kamien and Zang (1990) analyze an acquisition game among competing firms.
- Each firm:
  - Submits a bid for every other firm.
  - States an asking price for their own firm.
- The firm offering the highest bid above the asking price acquires the target.
- Key question: Can a Nash equilibrium lead to a monopoly where one firm acquires all competitors?

Key Findings

- A Nash equilibrium may not exist where one firm successfully acquires both competitors to form a monopoly.
- This outcome persists even if:
  - The resulting monopoly would generate higher profits.
  - All firms have incentives to engage in acquisition negotiations.
- The game's structure leads to challenges in aligning bids and asking prices across firms.

Kamien and Zang (1993)

- Extended their 1990 model by considering dynamic merger incentives over time.
- Focused on how market conditions evolve after initial mergers.
- showed that a profitable merger from oligopoly with three or more firms to monopoly may not be an equilibrium.
- Demonstrated that sequential mergers could reduce overall welfare, especially if cost synergies are minimal.

T Lindqvist and J Stennek (2005)

- Research Focus: Investigate why profitable mergers may not occur due to individual firms' incentives.
- Key Concept: The "insiders' dilemma" where firms prefer to remain outsiders to benefit more.

The Insiders' Dilemma

- Definition: A situation where a profitable merger fails because each firm prefers to stay out, anticipating higher individual gains.
- Theoretical Background: Based on models by Kamien and Zang (1990, 1993).

Experimental Design

- Objective: Test the insiders' dilemma hypothesis in a controlled laboratory setting.
- Methodology: Participants represent firms deciding on mergers with potential profitability.
- Variables:
  - Payoff structures.
  - Information availability.
  - Number of firms.

Key Findings

- Support for Insiders' Dilemma: Profitable mergers often did not occur due to firms' preference to remain outsiders.
- Fairness Considerations: Mergers requiring unequal surplus distribution were less likely to happen.
- Implications: Both strategic incentives and fairness perceptions influence merger decisions.

Nilssen and Sørgard (1998)

- Analyzed the impact of antitrust regulation on sequential mergers.
- Found that:
  - Strict antitrust policies can slow down or prevent waves of mergers.
  - Firms anticipate regulatory responses and adjust their merger strategies accordingly.

Gowrisankaran (1999)

- Developed a dynamic framework for sequential mergers.
- Each merger changes market structure, influencing the profitability of subsequent mergers.
- Key insight: Initial mergers may encourage or discourage further consolidation depending on efficiency gains and market power.

Inderst and Wey (2003)

- Focused on sequential mergers in differentiated product markets.
- Firms with similar products are more likely to merge in early stages.
- Mergers shift market dynamics, influencing the incentives for future consolidations.

Section II: Introduction to European MA Regulation

- Mergers and Acquisitions (MA) are a key focus of European competition law.
- Regulated primarily under the **EU Merger Regulation** (EUMR).
- The objective: Prevent concentrations that significantly impede effective competition.

Key Legal Texts on MA in the EU

- **Council Regulation (EC) No 139/2004** on the control of concentrations between undertakings.
  - Establishes the European Commission's role in merger control.
  - Defines thresholds for notification of mergers.
- **Council Regulation (EC) No 1/2003**
  - Implements competition rules under Articles 101 and 102 TFEU.

Current Guidelines and Policies

- Horizontal Merger Guidelines 2004
  - Assess impacts on competition from mergers of competitors.
  - Focus on market shares, concentration, and potential anti-competitive effects.
- Non-Horizontal Merger Guidelines (2008)
  - Evaluate vertical, conglomerate integration and foreclosure risks.
- Market Definition Notice 2024
  - Defines relevant product and geographic markets for competition analysis.

Key Objectives of EU Merger Control

- Preserve effective competition in the internal market.
- Prevent monopolies or dominant positions that harm consumers.
- Foster innovation and market efficiency.

Step 1: Notification

When is Notification Required?

- Mandatory for mergers exceeding EU thresholds:
  - Combined worldwide turnover exceeds €5 billion.
  - EU-wide turnover of at least two firms exceeds €250 million.
- Voluntary in cases with potential cross-border effects.

What Happens?

- Companies submit a formal notification to the European Commission.
- Initial information includes market share data, turnover, and market impact.

Step 2: Phase I Investigation

Objective:

- Conduct a preliminary assessment to identify potential concerns.

Key Activities:

- Review the relevant market and competitive dynamics.
- Collect input from stakeholders (e.g., competitors, suppliers, customers).
- Assess whether the merger raises significant competition concerns.

Outcome:

- **Clearance**: No concerns, the merger is approved.
- **Phase II Referral**: Further investigation required.

Definition of Relevant Market

What is the Relevant Market?

- The **relevant market** identifies the boundaries within which competition is analyzed for a merger.
- Two key dimensions:
  - **Product Market**: Substitutable goods or services based on demand and supply.
  - **Geographic Market**: The area where competitive conditions are sufficiently homogeneous.

Purpose:

- Assess whether a merger significantly impedes effective competition.
- Determine the competitive constraints on the merging parties.

Tools to Define the Relevant Market

SSNIP Test:

- *Small but Significant and Non-transitory Increase in Price*.
- Assesses whether consumers would switch to alternatives in response to a 5-10

Economic Evidence:

- Price correlation analysis.
- Cross-price elasticity of demand.

Market Surveys:

- Stakeholder feedback, including competitors, customers, and suppliers.

Additional Tools:

- Consumer preferences and purchasing patterns.
- Analysis of supply-side substitution (e.g., ability of firms to switch production quickly).

What is the SSNIP Test?

Definition:

- The *Small but Significant and Non-transitory Increase in Price* (SSNIP) test assesses whether a hypothetical price increase would lead consumers to switch to alternative products or services.
- Typically considers a price increase of **5-10%** over a sustained period.

Objective:

- Define the **relevant product market** and **geographic market**.
- Determine whether a product or region constitutes a separate market.

Key Concept:

- If enough consumers switch to substitutes, the price increase is unprofitable, and the market definition must expand.

How the SSNIP Test Works

Steps of the SSNIP Test:

1. Start with the smallest plausible market (e.g., a single product or brand).
2. Hypothesize a price increase of 5-10%.
3. Analyze consumer behavior:
    - Would consumers switch to other products or regions?
    - Would the price increase remain profitable?
4. If unprofitable: Expand the market definition and repeat.
5. Stop when a profitable price increase is possible.

Outcome:

- Defines the **smallest market** in which the firm has significant pricing power.

Concrete Example of the SSNIP Test

Scenario:

- A merger between two producers of **chocolate bars**.
- Initial market definition: Only **milk chocolate bars**.

Step 1: Apply a 5% Price Increase

- If many consumers switch to **dark chocolate bars**, the price increase becomes unprofitable.
- Result: Expand the product market to include dark chocolate.

Step 2: Test with Expanded Market

- Hypothesize a price increase across both milk and dark chocolate bars.
- If consumers switch to **other confectionery** (e.g., candy or biscuits), expand further.

Geographic Dimension:

- Start with a local region (e.g., one country).
- Test whether consumers would import chocolate from neighboring regions in response to a price increase.

Challenges with the SSNIP Test

Practical Limitations:

- **Data Availability:** Requires reliable data on consumer behavior and price elasticity.
- **Complex Substitution Patterns:** Difficult to predict substitution in multi-sided platforms or digital markets.
- **Dynamic Markets:** Markets with frequent innovation may render the test obsolete quickly.

Digital Market Issues:

- Products with zero prices (e.g., free apps) cannot be directly tested for price increases.
- Network effects complicate market boundaries.

Real-World Examples

Example 1: Microsoft/LinkedIn (2016)

- Issue: Whether professional networking sites constituted a separate market.
- Outcome: The SSNIP test was challenging due to network effects.

Example 2: Coca-Cola (2004)

- Issue: Whether Coca-Cola's soft drinks were in a distinct market from other beverages.
- Outcome: The SSNIP test helped assess substitutability with non-carbonated drinks.

Challenges in Defining the Relevant Market

Difficulties:

- Rapid market evolution, especially in digital sectors.
- Multi-sided platforms: How to define markets when different user groups interact (e.g., advertisers and users on Google).
- Data as a competitive asset: Hard to capture its value in traditional market analysis.
- Supply-side substitution: Difficulty in predicting how firms might adapt.

Potential Problems:

- **Over-Narrow Markets:** Excludes broader competitive constraints, leading to false positives.
- **Over-Broad Markets:** Misses potential dominance issues, leading to false negatives.

Examples of Market Definition Problems

Case 1: Facebook/WhatsApp (2014)

- Issue: Whether messaging apps and social media belonged to the same market.
- Outcome: Approved, but concerns about data concentration were raised later.

Case 2: Google Shopping (2017)

- Issue: Definition of online search and comparison shopping markets.
- Outcome: Found abuse of dominance in favoring its own services.

Case 3: Microsoft/LinkedIn (2016)

- Issue: Whether professional networking and social media were distinct markets.
- Outcome: Approved with conditions on data access and interoperability.

Key Updates in the Revised European Market Definition Notice

- Adopted by the European Commission on February 8, 2024.
- Reflects evolving market dynamics, particularly in digital and global contexts.
- Enhances legal certainty for antitrust market definitions.

Challenges in digital markets for the SSNIP Test

- SSNIP Test: Determines if a hypothetical price increase (5-10%) is profitable.
- Assumes stable demand and competitive substitutes.
- but in digital markets:
  - Free services (e.g., social media platforms).
  - Multi-sided interactions.
  - Non-price competition (e.g., data, quality).

Challenges in the Digital Economy

- **Multi-Sided Markets:**
  - Platforms serve multiple user groups (e.g., advertisers and consumers).
  - Interdependence of demand across sides.
- **Effects of Network Externalities:**
  - Value increases with user base size.
  - Market tipping towards a dominant player.
- **Data as a Competitive Factor:**
  - Access to large datasets strengthens market power.
  - Barriers to entry for competitors without equivalent data.

SSNIP Test in Digital Markets

- Free services make price increases irrelevant:
    - How to apply SSNIP without a price?
    - Alternative metrics: changes in quality, data access.
- Single-market focus misses cross-market effects:
    - Pricing decisions on one side impact the other.
    - Example: Ad pricing and consumer engagement.
- Need for a redefined framework for multi-sided markets.

Dynamic and Innovative Markets in the New notice

- Acknowledges the role of multi-sided platforms and digital ecosystems.
- Introduces the concept of system or dual markets.
- Advocates separate assessments for each side of a platform when needed.

Non-Price Competition Factors

- Highlights innovation, quality, and sustainability as key elements.
- Moves beyond traditional price-based competition metrics.
- Emphasizes their growing relevance in modern economies.

Types of Evidence for Market Definition

- Provides guidance on evidence such as:
  - Consumer preferences.
  - Price correlations and trade flows.
  - Insights from customers and competitors.
- Aims to enhance market analysis accuracy.

Platform Mergers: Key Issues

- **Start-Up Acquisitions:**
  - Potential elimination of future competitors.
  - Example: Facebook/Instagram, Google/Waze.
- **Consolidation of Market Power:**
  - Increased dominance through data accumulation.
  - Risk of "winner-takes-all" dynamics.
- **Innovation Suppression:**
  - Reduced incentives for new entrants.
  - Risk of "killer acquisitions."

Globalization and Digitalization in the New notice

- Adapts market definitions to reflect increased globalization.
- Recognizes the dynamic impact of digitalization on market structures.
- Ensures relevance in evolving economic environments.

Flexibility in Competition Assessments in theNew notice

- Offers accessible guidance with increased flexibility.
- Avoids rigid frameworks for defining markets.
- Supports adaptability in complex digital ecosystems.

Step 3: Phase II Investigation

Objective:

- Conducted by competition authorities (e.g., European Commission, FTC).
- Conduct an in-depth analysis of complex cases.
- Examine potential anti-competitive effects in greater detail.

Key Activities:

- Detailed market analysis, including economic modeling.
- Evaluation of remedies proposed by the merging parties.
- Stakeholder consultations for deeper insights.

Outcome:

- **Approval**: Unconditional or subject to remedies.
- **Prohibition**: Merger is blocked if anti-competitive effects cannot be mitigated.

Competitive Analysis

- Assessment of how the merger affects competition in relevant markets.
- Key areas of focus:
  - **Market shares:** Do the merging firms hold significant combined market power?
  - **Market concentration:** Measured using tools like the Herfindahl-Hirschman Index (HHI).
  - **Impact on rivals:** Will competitors be foreclosed or disadvantaged?
- **Goal**: Identify risks of creating or strengthening a dominant position.

Unilateral Effects Analysis

- **Definition:** Unilateral effects occur when a merger reduces competition, allowing the merged entity to profitably increase prices or reduce output without relying on competitors.
- Key considerations:
  - **Loss of direct competition:** Between merging firms in overlapping markets.
  - **Price increases:** Likelihood that consumers will face higher prices due to reduced rivalry.
- **Tests used:**
  - **UPP Test (Upward Pricing Pressure):** Evaluates whether the merger creates incentives to raise prices.
  - **GUPPI (Gross Upward Pricing Pressure Index):** Measures pricing incentives based on diverted sales and profit margins.

Upward Pricing Pressure (UPP) Test

- **Purpose:** Quantifies the incentives for the merged entity to increase prices post-merger.
- Key components:
  - **Diversion ratio:** Percentage of sales lost by one merging firm that is captured by the other.
  - **Profit margin:** Marginal profitability of diverted sales.
- **Formula:**

$$\text{UPP} = (\text{Diversion Ratio} \times \text{Profit Margin}) - \text{Efficiencies}$$

- **Interpretation:** A positive UPP indicates potential price increases and requires further investigation.

UPP Test Example

- **Scenario:** Firm A and Firm B are merging. Both sell similar products in the same market.
- **Data:**
  - Diversion Ratio from Firm A to Firm B: 20% (0.2).
  - Profit Margin of Firm B: 30% (0.3).
  - Expected Efficiencies: 5% (0.05).
- **Calculation:**

$$\text{UPP} = (0.2 \times 0.3) - 0.05$$

$$\text{UPP} = 0.06 - 0.05$$

$$\text{UPP} = 0.01$$

- **Result:** The positive UPP suggests there is upward pricing pressure, indicating potential price increases post-merger. Further analysis is needed.

Coordinated Effects Analysis

- **Definition:** Coordinated effects arise when a merger facilitates tacit collusion or coordination among remaining competitors.
- Conditions for coordinated effects:
  - **Transparency:** Competitors can easily monitor each other's behavior.
  - **Stability:** Market conditions support long-term coordination.
  - **Punishment mechanisms:** Ability to discipline firms that deviate from the coordinated outcome.
- **Concern:** Increased likelihood of higher prices, reduced output, or stifled innovation.

Barriers to Entry

- Analysis of how easily new competitors can enter the market.
- Types of barriers:
  - **Structural:** High capital costs, economies of scale, access to distribution channels.
  - **Regulatory:** Licensing requirements, compliance costs.
  - **Behavioral:** Predatory pricing, exclusivity agreements.
- **Concern:** High barriers may allow merged firms to sustain anti-competitive behavior.

Countervailing Buyer Power

- Assessment of whether powerful buyers can offset the merged firm's market power.
- **Key factors:**
  - Buyers' ability to switch suppliers.
  - Buyers' influence on price negotiations.
  - Availability of alternative sources of supply.
- **Outcome:** Strong buyer power may mitigate anti-competitive effects.

Efficiency Gains

- Mergers may produce efficiencies that benefit consumers:
  - Cost reductions (e.g., economies of scale, synergies).
  - Innovation leading to better products or services.
  - Enhanced competitiveness in global markets.
- **Key criterion: Efficiencies must be merger-specific and passed on to consumers.**

Remedies and Commitments

- If concerns are identified, remedies may be proposed to address them:
  - **Structural remedies:** Divestiture of overlapping business units.
  - **Behavioral remedies:** Ensuring access to essential facilities, non-discriminatory practices.
- **Goal:** Restore effective competition while allowing the merger to proceed.

Decision Outcomes

- **Approval:** Merger is cleared without conditions.
- **Conditional clearance:** Merger is approved with remedies to address concerns.
- **Prohibition:** Merger is blocked due to unresolved competition issues.
- **Appeals:** Parties may challenge the decision in court.

What Are Behavioral Remedies?

Definition

Behavioral remedies are commitments by merging parties to modify their conduct post-merger to preserve competition and prevent anti-competitive practices.

Key Characteristics:

- Focus on regulating future behavior.
- Aim to ensure continued market access and prevent abuse of market power.
- Often used in cases where structural remedies (e.g., divestitures) are not feasible.

Legal Basis:

- Article 8(2) of the EU Merger Regulation.

Examples of Behavioral Remedies

- **Non-Discrimination Commitments:**
  - Example: **Google/Fitbit (2020)**
  - Remedy: Google committed not to use Fitbit health data for targeted advertising in the EU.
  - Issue: Concerns over enforcement and data usage monitoring.
- **Access to Infrastructure:**
  - Example: **Deutsche Börse/London Stock Exchange (2017)**
  - Remedy: Commitments to ensure third-party access to clearing services.
  - Issue: Complexity in ensuring fair and equal access.
- **Interoperability Requirements:**
  - Example: **Microsoft/LinkedIn (2016)**
  - Remedy: Microsoft committed to ensure interoperability between LinkedIn and competing professional networks.
  - Issue: Difficulty in verifying compliance.

Challenges with Behavioral Remedies

- **Monitoring and Enforcement:**
  - Continuous oversight required by regulators.
  - High cost of compliance and risk of circumvention.
- **Complexity:**
  - Behavioral remedies can be highly technical and difficult to implement effectively.
- **Limited Effectiveness:**
  - Remedies may fail to fully address anti-competitive concerns.
  - Example: **Google/DoubleClick (2008)**: Despite commitments, concerns about market dominance in digital advertising persisted.
- **Unintended Consequences:**
  - Potential distortion of market dynamics or harm to innovation.

Benefits of Behavioral Remedies

- **Flexibility:**
  - Allows firms to maintain synergies from mergers while addressing competition concerns.
- **Preserves Market Structure:**
  - Avoids the need for divestitures that could disrupt the market.
- **Tailored Solutions:**
  - Remedies can be specifically designed for complex industries (e.g., tech and telecom).

Example: Siemens/Alstom (2019)

- Merger blocked after a Phase II investigation.
- Concerns over reduced competition in rail signaling and high-speed trains.

Background of the Alstom-Siemens Merger

- Proposed merger between Alstom (France) and Siemens Mobility (Germany).
- Aim: Create a European rail giant to compete with global players like CRRC (China).
- European Commission blocked the merger in February 2019.
- Reason: Concerns over reduced competition in the high-speed rail and signaling markets.

Criticism 1: Failure to Consider Global Competition

- Critics argue the decision focused narrowly on the European market.
- Ignored competitive pressure from global giants like CRRC.
- Fear of undermining Europe's ability to create "European champions."
- **Quote:** "Europe needs to compete on a global scale, not just within its borders."

Criticism 2: Impact on Innovation

- Merger could have pooled resources to drive innovation in rail technology.
- Blocking the deal may have limited potential advancements in green and high-speed transport.
- **Criticism:** Decision prioritizes short-term market concerns over long-term innovation.

Criticism 3: Rigid Application of Competition Rules

- Critics say the Commission applied competition rules too rigidly.
- Limited consideration of strategic industrial policies.
- **Example:** Other regions (e.g., China, USA) prioritize industrial strength over strict competition.

Criticism 4: Missed Opportunity for European Unity

- Merger could have symbolized European industrial cooperation.
- Blocking it sent mixed signals about the EU's commitment to global competitiveness.
- **Criticism:** Decision risks fragmenting European industrial strategy.

Other Open Questions:

- Digital markets and innovation require adaptations to traditional tools.
- How should merger thresholds adapt to capture digital market realities?
- What role should data and innovation play in future assessments?

Key Competition Issues in Digital MA

- **Market Concentration:**
  - Dominant platforms acquiring rivals to reinforce their position.
  - Example: Facebook's acquisitions of Instagram and WhatsApp.
- **Foreclosure Risks:**
  - Preventing competitors from accessing critical resources (e.g., data).
  - Example: Google/DoubleClick and advertising markets.
- **Killer Acquisitions:**
  - Buying innovative start-ups to suppress potential threats.
  - Example: Amazon's acquisition of Diapers.com.
- **Impact on Innovation:**
  - Reduced incentives for innovation post-merger.
  - Example: Qualcomm/NXP and chip innovation concerns.

Examples of Digital MA Cases

- **Facebook/Instagram (2012):**
    - Concern: Elimination of a growing competitor.
    - Outcome: Approved without remedies; later seen as an enabler of Facebook's dominance.
    - now criticized for suppressing a potential rival.
- **Google/Fitbit (2021):**
    - Concern: Access to health data and integration with Google's ecosystem.
    - Outcome: Approved with commitments on data use.
- **Microsoft/LinkedIn (2016):**
    - Concern: Integration of LinkedIn data into Microsoft products.
    - Outcome: Approved with behavioral remedies.
- **Amazon/Deliveroo (2020):**
    - Concern: Market foreclosure in food delivery.
    - Outcome: Approved after an extensive investigation.

Risks in Digital MA

- **Barriers to Entry:**
  - Large platforms create "winner-takes-all" dynamics.
  - High costs for new entrants to compete.
- **Data Monopolies:**
  - Exclusive access to valuable datasets.
  - Example: Facebook's user data vs. competitors.
- **Consumer Harm:**
  - Reduced choice and innovation.
  - Example: Google's practices in Android restricting competitors.
- **Regulatory Challenges:**
  - Difficulty in identifying anti-competitive risks early.
  - Underestimation of long-term effects.

Recommendations for Regulators

- **Broaden Merger Review Thresholds:**
  - Consider value transactions below turnover thresholds (e.g., start-ups).
  - Germany and Austria now include transaction value.
  - Captures acquisitions of start-ups with low revenues but high strategic importance.
- **Focus on Data and Ecosystems:**
  - Analyze data access and ecosystem impacts.
- **Strengthen Remedies:**
  - Behavioral and structural remedies to preserve competition.
- **Monitor Innovation Dynamics:**
  - Ensure start-ups have room to grow without fear of suppression.

Multi-Sided Platforms and Ecosystems

- **Multi-Sided Markets:**
  - Analyzing cross-group effects is crucial.
  - Example: Microsoft/LinkedIn (2016) – impact on professional networks and advertising.
- **Ecosystem Effects:**
  - Digital giants operate interconnected services.
  - Analysis must consider cumulative market power across ecosystems.
  - Example: Google's integration of services like Search, YouTube, and Android.

Why Amazon Is Not a Single-Market Firm?

- Amazon operates simultaneously in:
  - E-commerce,
  - Cloud computing (AWS),
  - Digital advertising,
  - Logistics and delivery,
  - Media and entertainment,
  - Smart devices and data-driven services.
- These activities are **economically interconnected**.
- Amazon should be analyzed as an **ecosystem**, not as a standalone retailer.

Economic Definition of a Digital Ecosystem

- A digital ecosystem is a set of complementary services:
  - Connected through data flows,
  - Reinforced by network effects,
  - Sustained by economies of scope.
- Market power arises from:
  - Cross-market leverage,
  - User lock-in,
  - Cumulative advantages.

The Core: Marketplace and Prime

- Amazon Marketplace connects:
  - Consumers,
  - Third-party sellers,
  - Brands,
  - Advertisers.
- Amazon Prime bundles:
  - Fast delivery,
  - Video and music,
  - Exclusive services.
- **Economic effect:**
  - High switching costs,
  - Strong consumer lock-in.

AWS: The Financial Backbone

- Amazon Web Services (AWS):
  - Cloud infrastructure,
  - AI, data storage, APIs.
- AWS generates:
  - A large share of Amazon's operating profits.
- **Economic implication:**
  - Cross-subsidization of retail and logistics,
  - Investment capacity unmatched by rivals.

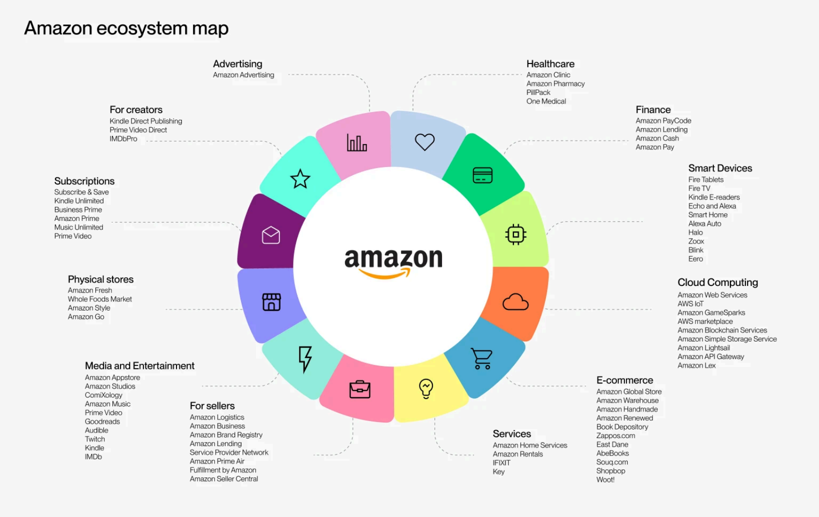
Logistics and Fulfillment as Entry Barriers

- Fulfillment by Amazon (FBA):
  - Warehousing,
  - Last-mile delivery,
  - Inventory management.
- Sellers become:
  - Operationally dependent on Amazon.
- **Barrier to entry:**
  - Competitors must replicate an entire logistics system,
  - Not just a digital platform.

Data and Advertising Feedback Loop

- Amazon collects:
  - Search data,
  - Purchase data,
  - Pricing and demand data.
- Amazon Advertising:
  - Sellers pay for visibility,
  - Ads are directly linked to transactions.
- **Economic effect:**
  - Sellers are simultaneously:
    - Customers,
    - Business partners,
    - Advertisers.

Amazon Ecosystem Map



Amazon integrates commerce, cloud, data, logistics, media, and devices into a single ecosystem.

Multi-Market Power and Ecosystem Leverage

- Market power does not stem from a single dominant position.
- Instead, it arises from:
  - Reinforcing positions across markets,
  - Data-driven complementarities,
  - Ecosystem lock-in.
- Traditional tools (HHI, SSNIP) struggle to capture this structure.

Implications for Competition Policy

- Risks are often **non-price**:
  - Foreclosure of sellers,
  - Self-preferencing,
  - Data concentration.
- Harm may appear without:
  - Price increases,
  - Output reductions.
- Calls for:
  - Ex ante regulation (DMA),
  - Ecosystem-wide analysis.

Amazon's M&A Strategy Through an Ecosystem Lens

- Acquisitions target:
  - Strategic complements,
  - Data-enhancing activities,
  - Infrastructure expansion.
- Examples:
  - Whole Foods: offline data and logistics,
  - MGM: content for Prime Video,
  - iRobot (attempted): household data and smart home integration.
- **Goal:** Strengthen ecosystem dominance rather than market share alone.

Key Takeaways

- Amazon exemplifies ecosystem-based market power.
- Competitive advantages are cumulative and self-reinforcing.
- Merger control and competition policy must:
  - Go beyond single-market analysis,
  - Focus on data, leverage, and lock-in.

Final Message

Amazon does not compete in one market. It competes through an ecosystem that reshapes competition itself.

Advanced Analytical Tools

- **Market Simulations:**
  - Simulate price, quality, and innovation impacts post-merger.
- **Big Data Analysis:**
  - Use of large datasets to understand user behavior and market dynamics.
- **Qualitative Tests:**
  - Go beyond SSNIP tests to assess quality and innovation.

Cooperation and Post-Merger Monitoring

- **International Cooperation:**
  - Collaboration between regulators on cross-border mergers.
  - Example: Meta/Kustomer investigated by EU and US regulators.
- **Post-Merger Monitoring:**
  - Follow-up on commitments made during merger approval.
  - Example: Google/Fitbit – monitoring data usage compliance.
- **Revisiting Past Mergers:**
  - Reviewing impacts of earlier decisions (e.g., Facebook/Instagram).

Cross-Border Mergers

- Cross-border mergers involve multinational companies operating in multiple jurisdictions.
- Regulators assess potential competition impacts within their own legal frameworks.
- Cooperation between regulators is essential to ensure consistency and avoid conflicts.

Cooperation in Cross-Border Mergers

- **Information Sharing:**
  - Exchange of non-confidential data about market dynamics and evidence.
- **Harmonization of Remedies:**
  - Aligning commitments such as divestitures to address global concerns.
- **Bilateral and Multilateral Agreements:**
  - Bilateral agreements between regulators (e.g., EU-U.S. agreements).
  - Platforms like the International Competition Network (ICN).
- **Coordinated Timelines:**
  - Aligning review schedules to minimize delays and ensure consistency.

Benefits and Challenges of Cooperation

Benefits:

- Reduces regulatory duplication and enhances efficiency.
- Promotes consistent decisions across jurisdictions.
- Facilitates fair competition in global markets.

Challenges:

- Differences in legal frameworks and enforcement priorities.
- Variations in local market conditions and economic factors.
- Geopolitical considerations affecting decision-making.

GE/Honeywell (2001)

Context: General Electric attempted to acquire Honeywell.
Outcome:

- **U.S. Approval:** Focused on horizontal competition.
- **EU Rejection:** Concerned about portfolio effects and bundling.

Reason for Divergence:

- Different emphasis on vertical and conglomerate effects in the EU.

Boeing/McDonnell Douglas (1997)

Context: Boeing sought to acquire McDonnell Douglas.
Outcome:

- **U.S. Approval:** McDonnell Douglas was no longer a strong competitor.
- **EU Approval with Reservations:** Concerns over global dominance and contracts.

Reason for Divergence:

- EU's focus on long-term dominance and exclusive contracts.

Microsoft/Activision Blizzard (2023)

Context: Microsoft's acquisition of Activision Blizzard.
Outcome:

- **UK CMA Block:** Concerns about cloud gaming dominance.
- **EU Approval:** Conditional on licensing commitments.
- **U.S. Review:** Ongoing FTC lawsuit.

Reason for Divergence:

- Different focus on emerging cloud gaming markets.

Qualcomm/NXP (2018)

Context: Qualcomm sought to acquire NXP Semiconductors.
Outcome:

- **Most Jurisdictions Approved:** Commitments addressed concerns.
- **China Blocked:** Led to abandonment of the merger.

Qualcomm: Nationality and Core Activities

- **Nationality:** United States
- **Headquarters:** San Diego, California
- **Core activities:**
  - Semiconductor design (fabless model)
  - Mobile communication technologies (3G, 4G, 5G)
  - System-on-chip (SoC) for smartphones
  - Extensive portfolio of standard-essential patents (SEPs)
- **Business model:**
  - Chip sales
  - Licensing of intellectual property (royalties)

NXP Semiconductors: Nationality and Core Activities

- **Nationality:** European (Netherlands)
- **Legal seat:** Netherlands
- **Origins:** Spin-off from Philips Semiconductors
- **Core activities:**
  - Automotive semiconductors
  - Internet of Things (IoT)
  - Secure elements and Near Field Communication (NFC)
  - Chips for payments, identification and industrial uses
- **Strategic position:**
  - Strong presence in fast-growing and innovation-intensive markets

The Qualcomm / NXP Merger Proposal

- **Announcement:** October 2016
- **Transaction value:** Approximately $44 billion
- **Nature of the transaction:**
  - Acquisition of NXP by Qualcomm
  - Cross-border (US–EU) merger
- **Strategic objectives:**
  - Diversification beyond smartphones
  - Strengthening position in automotive and IoT markets
  - Combining connectivity and secure chip technologies

Economic Rationale of the Merger

- Expected **efficiency gains**:
  - R&D complementarities
  - Economies of scope
  - Technological synergies
- Creation of a **leading global semiconductor ecosystem**
- Positioning for future technologies:
  - 5G
  - Autonomous vehicles
  - Connected devices

Authorities Reviewing the Merger

- United States: Federal Trade Commission (FTC)
- European Union: European Commission (DG Competition)
- China: MOFCOM (later SAMR)
- Other jurisdictions also reviewed the merger

United States (FTC): Approval

- The FTC approved the merger in 2018
- Main focus:
    - Traditional competition analysis
    - Horizontal overlaps in specific chip markets
- Innovation concerns:
    - Considered but not decisive
    - Efficiency and R&D synergies emphasized
- Overall approach:
    - Relatively permissive
    - High burden of proof for innovation harm

European Union: Approval with Remedies

- Clearance granted in January 2018, subject to commitments
- Identified concerns:
  - NFC chips
  - Secure elements
  - Potential foreclosure of rivals
- Innovation-related issues:
  - Risk of reduced innovation diversity
  - Restricted access to key technologies
- Remedies imposed:
  - FRAND licensing commitments
  - Guarantees of interoperability

China (MOFCOM / SAMR): Non-Approval

- China did not approve the merger before the deadline
- Official concerns:
    - Strengthening of Qualcomm's market power
    - Impact on innovation in semiconductors
    - Access to key technologies for Chinese firms
- Broader context:
    - Strategic importance of semiconductors
    - US–China trade tensions
    - Industrial policy considerations

Why the Merger Ultimately Failed

- Merger control is cumulative:
  - Approval required in all major jurisdictions
- China as a key market:
  - Major semiconductor demand
  - Critical manufacturing and assembly hub
- Without Chinese approval:
  - Legal and economic implementation impossible
- Qualcomm abandoned the merger in July 2018

Key Lessons from Qualcomm / NXP

- Innovation is central in high-tech merger analysis
- Authorities may diverge significantly in their assessments
- Competition policy increasingly intersects with:
    - Industrial policy
    - Geopolitics
    - Strategic technologies
- The case illustrates limits to global convergence in merger control

Conclusion

- Qualcomm / NXP is a landmark case in digital and high-tech merger control
- Highlights the role of innovation and ecosystem effects
- Shows how a single jurisdiction can block a global merger

Transition: Digital Markets and the DMA

- Growing dominance of digital platforms in global markets.
- Key concerns: abuse of market power, anti-competitive practices.
- Focus on EU antitrust actions against:
  - Apple
  - Amazon
  - Google
- Regulatory framework: Digital Markets Act (DMA) and Digital Services Act (DSA).

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Chapter III: Abuse of Dominant Position and Competition Issues in the Digital Economy

Focus on EU Policies and the Digital Markets Act (DMA)

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Definition of Abuse of Dominant Position

Definition (Article 102 TFEU)

Abuse of dominant position occurs when a company holding a dominant market position engages in practices that distort competition, harm consumers, or unfairly exclude competitors.

Key Elements:

- **Dominant Position:** A position of economic strength allowing a firm to behave independently of competitors and consumers.
- **Abuse:** Exploitative or exclusionary conduct such as:
  - Predatory pricing
  - Tying and bundling
  - Refusal to supply

Legal Framework: Governed under Article 102 of the Treaty on the Functioning of the European Union (TFEU).

Introduction

- In 2018, the European Commission fined Google for abusing its dominant position in the mobile operating systems market.
- The case focused on illegal **tying and bundling** practices involving Google's Android operating system.

Key Facts of the Case

- Google required manufacturers using Android to:
  - Preinstall Google Search as the default search engine.
  - Preinstall Google Chrome as the default browser.
  - Include Google Play Store to access key apps.
- These requirements were conditions for granting licenses to the Android operating system.

The European Commission's Decision

- The Commission ruled that Google's practices:
  - Restricted competition in the search and browser markets.
  - Limited consumer choice by discouraging alternatives.
  - Strengthened Google's dominance in online search.
- Google was fined a record **€4.34 billion**.

Implications of the Decision

- Google was required to:
  - Stop enforcing exclusivity agreements.
  - Allow manufacturers to preinstall competing apps.
  - Offer an *"app choice screen"* to users in the European Economic Area (EEA).
- The case set a precedent for antitrust enforcement in digital markets.

Key Competition Issues in the Digital Economy

- **Self-Preferencing:**
  - Example: Amazon promoting its own products over third-party sellers.
- **Data Exploitation:**
  - Example: Platforms collecting and leveraging user data for competitive advantage.
- **Network Effects:**
  - Large user bases lock in consumers, making it difficult for competitors to enter the market.
- **Gatekeeping Power:**
  - Dominant platforms act as gatekeepers, controlling access to markets and consumers.
  - Example: Apple's App Store rules.
- **Killer Acquisitions:**
  - Example: Big Tech acquiring innovative start-ups to eliminate potential competition (e.g., Facebook and Instagram).

Tying and Bundling example

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What is Self-Preferencing?

Definition

Self-preferencing occurs when a company favors its own products or services over those of competitors on a platform it controls, reducing fair competition.

Key Characteristics:

- Manipulation of search rankings to prioritize in-house products.
- Exclusive access to platform data to improve competitive advantage.
- Discrimination against third-party products.

Impact:

- Harms competition by reducing visibility of rivals' products.
- Limits consumer choice.

Examples of Self-Preferencing

- **Google Search:**
  - Favoring Google Shopping results over competing services.
  - EU fine: €2.42 billion in 2017 for anticompetitive practices.
- **Apple App Store:**
  - Prioritizing its own apps like Apple Music over competitors like Spotify.
- **Amazon Marketplace:**
  - Promoting Amazon-branded products (e.g., Amazon Basics) above third-party sellers in search results.

Challenges Specific to Digital Platforms

- **Algorithmic Bias:**
  - Lack of transparency in how search results or rankings are determined.
- **Market Foreclosure:**
  - Blocking competitors' access to critical markets or data.
- **Price Discrimination:**
  - Using data to charge different prices based on user profiles.
- **Lack of Interoperability:**
  - Restricting third-party applications or services from interacting with dominant platforms.
- **Consumer Harm:**
  - Reduced innovation and choice due to anti-competitive practices.

Digital Markets Act (DMA)

Overview:

- Objectives: Fairness and contestability in digital markets.
- Key provisions: Gatekeeper obligations.

Discussion:

- How does the DMA complement traditional competition policy?

What is a Gatekeeper?

Definition (DMA Article 3)

A "Gatekeeper" is a company that controls access to digital markets, acting as an intermediary between businesses and consumers due to its strong economic position, large user base, and entrenched presence in the market.

Criteria to Qualify as a Gatekeeper:

- Annual turnover in the EU of at least €7.5 billion or market capitalization of €75 billion.
- Core platform services with at least 45 million monthly active EU users and 10,000 annual active business users.
- Significant, entrenched, and durable market power.

Examples of Gatekeepers

Core Platform Services Identified:

- **Search Engines:** Google.
- **Social Networks:** Meta (Facebook, Instagram, WhatsApp).
- **E-commerce Platforms:** Amazon.
- **Operating Systems:** Apple (iOS), Google (Android).
- **Online Advertising Services:** Google Ads.
- **App Stores:** Apple App Store, Google Play Store.

Case Studies:

- **Apple App Store:** Controls app distribution and payment processing for iOS devices.
- **Amazon Marketplace:** Acts as both a platform operator and competitor to third-party sellers.

Key Obligations for Gatekeepers Under the DMA

Obligations to Ensure Fair Competition:

- **Data Sharing:** Gatekeepers must allow businesses to access data generated on their platforms.
- **Interoperability:** Ensure compatibility with third-party services (e.g., messaging apps).
- **Platform Neutrality:** Avoid favoring their own services over competitors'.
- **Transparency:** Provide clear terms for business users and report compliance with obligations.

Prohibited Practices:

- Using non-public data from business users to compete with them (e.g., Amazon Marketplace).
- Restricting app developers from using alternative payment systems (e.g., Apple App Store).
- Bundling services to lock users into their ecosystem.

Enforcement of the DMA

Monitoring by the European Commission:

- Gatekeepers must submit annual compliance reports.
- Non-compliance can result in fines up to 10% of global turnover.

Penalties:

- **Fines:** Up to 20% of global turnover for repeated violations.
- **Structural Remedies:** Break-up of companies as a last resort.

Challenges:

- Balancing regulation with innovation.
- Ensuring robust enforcement across Member States.

Implications for Gatekeepers

Opportunities:

- Increased trust from business users and consumers.
- Potential for innovation in response to regulatory requirements.

Challenges:

- Higher compliance costs and operational changes.
- Potential legal disputes over interpretation of the DMA.

Emerging Topics:

- Sustainability and competition.
- Vertical integration in the digital economy.

Vertical integration in the digital economy

- **Vertical integration**: Control over multiple stages of production or supply chain.
- **Digital sector**: Involves hardware, software, content, and services.
- **Examples**: Companies like Amazon, Apple, Google, and Netflix.

Benefits of Vertical Integration

- **Control over supply chain**: Improved efficiency and cost savings.
- **Data utilization**: Access to vast amounts of user data to optimize services.
- **Innovation**: Seamless integration of hardware, software, and services.
- **Network effects**: Amplifies user engagement and market dominance.

Examples of Vertical Integration

- **Amazon**: E-commerce, logistics, AWS, media streaming.
- **Apple**: Hardware (iPhones), software (iOS), services (App Store).
- **Google**: Search engine, Android, YouTube, advertising.
- **Netflix**: Content production and streaming platform.

Impact on Competition

- **Increased market power**: Dominance across multiple sectors.
- **Barriers to entry**: Harder for smaller firms to compete.
- **Anti-competitive practices**:
  - Tying products and services.
  - Predatory pricing.
  - Exclusive deals.

Consumer Welfare

- **Benefits to consumers**:
  - Lower prices.
  - Better-quality products.
  - Seamless user experiences.
- **Concerns for consumers**:
  - Reduced choice.
  - Risk of innovation stifling.

Regulatory Challenges

- **Antitrust concerns**: Scrutiny of mergers and acquisitions.
- **Potential for breakups**: Calls for splitting dominant firms.
- **Balancing benefits and fairness**: Encouraging innovation while preventing market abuse.

Discussion:

- Role of interoperability and Data portability in promoting competition.

Data Portability

Definition:

- The ability for a user to transfer their data from one service (A) to another (B).
- Supported by GDPR as an individual right.

Examples:

- List of songs listened to, calendar entries, notes.

Implications for Competition:

- Reduces data lock-in, facilitating user switching.
- Promotes access to anonymized and aggregated data.
- Risks: Minimal anti-competitive concerns under GDPR.

Protocol Interoperability

Definition:

- Ensures that systems can fully interact to provide complementary services.
- Distinct from data interoperability (see next slide).

Examples:

- Operating systems, phone charging protocols.

Implications for Competition:

- Enables complementary services and competition on merit.
- May require standard-setting, with both benefits and risks.

Data Interoperability

Definition:

- Real-time, potentially standardized access to data via APIs.

Examples:

- Add-ons for platforms (e.g., Slack, Gmail), IoT data access.

Challenges:

- Security risks (e.g., Cambridge Analytica scandal).
- Need for technical and legal standards.

Implications for Competition:

- Facilitates multi-homing and complementary services.
- Potential risks of anti-competitive information exchange.

Full Protocol Interoperability

Definition:

- Enables interaction between multiple substitute services.
- Requires deeper standardization than protocol interoperability.

Examples:

- Messaging systems, mobile phone networks, email protocols.

Implications for Competition:

- Reduces network effect-driven lock-in.
- Risks: Potential for reduced innovation and collusion.

From Market Power to Behavioral Power

Key idea

Even when alternatives exist, users may not switch because:

- Cognitive effort is costly
- Defaults and framing guide attention
- Frictions discourage active choice

From Market Power to Behavioral Power

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- Defaults and framing guide attention
- Frictions discourage active choice

Regulatory implication (DMA)

Competition may be **theoretically possible** but **behaviorally ineffective**.

1. Default Bias / Status Quo Bias

- Users tend to stick with pre-selected options
- Changing defaults requires effort and perceived risk

Interface examples

- Default search engine or browser
- Pre-installed apps
- Default privacy or consent settings

Effect

Defaults strongly influence market shares and user behavior.

2. Friction Costs and Sludge

- Frictions = small obstacles that delay or discourage action
- Sludge = excessive or asymmetric frictions

Interface examples

- Easy sign-up, complex cancellation
- Multiple screens to opt out
- Hidden unsubscribe buttons

Behavioral mechanism

Procrastination, attention scarcity, decision fatigue

3. Salience and Position Bias

- Users focus on what is most visible
- Items placed first are perceived as better or safer

Interface examples

- Ranking in search results
- Highlighted or "recommended" options
- Featured apps or products

Strategic use

Self-preferencing by gatekeepers through interface prominence

4. Framing Effects

- Choices depend on how options are presented
- Positive vs. negative framing changes decisions

Interface examples

- "Improve your experience" vs. "Limit personalization"
- "Accept all" vs. "Manage settings"
- Emotional or reassuring language

Risk

Users may consent without understanding real consequences.

5. Cognitive Load and Information Overload

- Too much information reduces decision quality
- Users rely on heuristics or shortcuts

Interface examples

- Long privacy policies
- Complex menus and nested options
- Excessive legal or technical language

Outcome

Rational ignorance and passive acceptance of defaults.

6. Switching Costs and Learning Costs

- Switching platforms requires time and learning
- Users fear losing data, habits, or compatibility

Interface examples

- Ecosystem lock-in (apps, cloud, accounts)
- Non-interoperable services
- Complex data portability

Behavioral effect

Inertia even when better alternatives exist.

7. Trust and Authority Bias

- Users trust what appears official or system-integrated
- Platform branding signals safety and quality

Interface examples

- Native apps vs. third-party apps
- System warnings favoring default services
- Platform-endorsed recommendations

Implication

Gatekeepers benefit from perceived legitimacy.

Summary: Biases Activated by Interfaces

- Default bias
- Friction costs / Sludge
- Salience and position bias
- Framing effects
- Cognitive overload
- Switching and learning costs
- Trust and authority bias

Why This Matters for Competition Policy

Key insight

Market power is reinforced by **behavioral power**.

- Formal choice is not sufficient
- Effective choice requires neutral interfaces
- DMA focuses on:
  - Choice screens
  - Symmetric user journeys
  - Default neutrality

Conclusion

- Interfaces shape behavior as much as prices or quality
- Gatekeepers can exploit predictable biases
- Behavioral economics is essential for digital regulation

Competition is not only a market issue, but also a cognitive one.